

Chapter 1

Connor Mooneyhan

April 9, 2024

Exercise 1. A nonempty family of sets $\mathcal{R} \in \mathcal{P}(X)$ is called a **ring** if it is closed under finite unions and differences (i.e., if $E_1, \dots, E_n \in \mathcal{R}$, then $\cup_1^n E_j \in \mathcal{R}$, and if $E, F \in \mathcal{R}$, then $E \setminus F \in \mathcal{R}$). A ring that is closed under countable unions is called a **σ -ring**.

- a. Rings (resp. σ -rings) are closed under finite (resp. countable) intersections.
- b. If $\mathcal{R}$ is a ring (resp. σ -ring), then $\mathcal{R}$ is an algebra (resp. σ -algebra) iff $X \in \mathcal{R}$.
- c. If $\mathcal{R}$ is a σ -ring, then $\{E \subset X : E \in \mathcal{R} \text{ or } E^c \in \mathcal{R}\}$ is a σ -algebra.
- d. If $\mathcal{R}$ is a σ -ring, then $\{E \subset X : E \cap F \in \mathcal{R} \text{ for all } F \in \mathcal{R}\}$ is a σ -algebra.

Proof.

- a. If $\mathcal{R}$ is a σ -ring and $\{E_n\}_{n \in \mathbb{N}} \subset \mathcal{R}$, then

$$\bigcap_{n \in \mathbb{N}} E_n = \left(\bigcup_{n \in \mathbb{N}} E_n \right) \setminus \left(\bigcup \{E_a \setminus E_b : a, b \in \mathbb{N}\} \right) \quad (1)$$

Considering the left and right components of the difference in the right hand side of (1), the left is clearly a countable union of members of $\mathcal{R}$, and the right can be seen to be a countable union by identifying each $E_a \setminus E_b$ with $\frac{a}{b}$. Thus the right hand side of (1) is a difference of countable unions of members of $\mathcal{R}$, so it is in $\mathcal{R}$.

If $\mathcal{R}$ is a ring, the argument above applies by making $\{E_n\}$ a finite set.

- b. In either case, if $\mathcal{R}$ is a ring (resp. σ -ring) and an algebra (resp. σ -algebra), then given $E \in \mathcal{R}$, $X = E \cup E^c \in \mathcal{R}$.

Conversely, if $\mathcal{R}$ is a ring (resp. σ -ring) and $X \in \mathcal{R}$, then given $E \in \mathcal{R}$, $E^c = X \setminus E \in \mathcal{R}$, so $\mathcal{R}$ is closed under complements. $\mathcal{R}$ is already closed under finite (resp. countable) unions, so $\mathcal{R}$ is an algebra.

- c. Let $A = \{E \subset X : E \in \mathcal{R} \text{ or } E^c \in \mathcal{R}\}$. Given $E \in A$, by definition $E^c \in A$ since $(E^c)^c = E \in A$.

Let $\{E_n\}_{n \in \mathbb{N}} \subset A$, and express this set as the union of two sets, namely

$$\{E_n\}_{n \in \mathbb{N}} = \{F_j\}_{j \in J} \cup \{G_k\}_{k \in K}$$

where $\{F_j\}_{j \in J} \subset A$ is the collection of all sets in A who are members of $\mathcal{R}$ and $\{G_k\}_{k \in K} \subset A$ is the collection of sets in A who are not themselves in $\mathcal{R}$, but whose complements are. Then

$$\begin{aligned} \bigcup_{n \in \mathbb{N}} E_n &= \bigcup_{j \in J} F_j \cup \bigcup_{k \in K} G_k \\ &= \bigcup_{j \in J} F_j \cup \left(\bigcap_{k \in K} G_k^c \right)^c \end{aligned} \quad (1)$$

Since each F_j is in $\mathcal{R}$, $\bigcup_{j \in J} F_j \in \mathcal{R} \subset A$. Also, we know from (a) that σ -rings are closed under countable intersections, so since $G_k^c \in \mathcal{R}$ for each G_k , we have $\bigcap_{k \in K} G_k^c \in \mathcal{R}$. Its complement is also in $\mathcal{R}$ since $\mathcal{R}$ is closed under complementation. Thus (2) is a union of two members of $\mathcal{R}$, and is therefore in $\mathcal{R} \subset A$.

- d. Let $A = \{E \subset X : E \cap F \in \mathcal{R} \text{ for all } F \in \mathcal{R}\}$. Note that $\mathcal{R} \subset A$ since $\mathcal{R}$ is closed under finite intersections. Given any $E, F \in \mathcal{R}$, we have

$$E^c \cap F = (X \setminus E) \cap F = F \setminus E$$

which is a difference in $\mathcal{R}$ and is thus in $\mathcal{R} \subset A$.

Given $\{E_j\}_{j \in \mathbb{N}} \subset A$ and arbitrary $F \in A$,

$$\left(\bigcup_{j \in \mathbb{N}} E_j \right) \cap F = \bigcup_{j \in \mathbb{N}} (E_j \cap F),$$

which is a countable union of elements in $\mathcal{R}$. Thus since the choice of F was arbitrary, $\bigcup \{E_j\} \in A$.

□

Exercise 2. Complete the proof of Proposition 1.2.

Proof. The construction of open intervals from half-open intervals and rays is standard. □

Exercise 3. Let $\mathcal{M}$ be an infinite σ -algebra.

- a. $\mathcal{M}$ contains an infinite sequence of disjoint nonempty sets.
- b. $\text{card}(\mathcal{M}) \geq c$.

Proof.

- a. Choose a sequence $\{E_n\}_{n \in \mathbb{N}}$ of nonempty sets in $\mathcal{M}$ (this requires the axiom of choice). We then derive the sequence $\{F_n\}$ defined recursively by $F_1 = E_1$ and $F_j = E_j \setminus F_{j-1}$ when $j \geq 2$. These are disjoint by construction and are all in $\mathcal{M}$ since $\mathcal{M}$ is closed under differences.
- b. Choose a sequence $\{E_j\}_1^\infty$ of disjoint nonempty sets in $\mathcal{M}$ (allowable by part (a) of this exercise). Define $f : \mathcal{P}(\mathbb{N}) \rightarrow \mathcal{M}$ by $f : S \mapsto \bigcup_{k \in S} E_k$. This is well-defined since $f(S)$ is a countable union of members of $\mathcal{M}$, so $f(S) \in \mathcal{M}$. Also, f is an injection since all the E_j are disjoint. This brings us to the conclusion that since $\text{im } f \subset \mathcal{M}$,

$$\text{card } \mathcal{M} \geq \text{card}(\text{im } f) = \text{card}(\mathcal{P}(\mathbb{N})) = c.$$

□

Exercise 4. An algebra $\mathcal{A}$ is a σ -algebra iff $\mathcal{A}$ is closed under countable increasing unions (i.e., if $\{E_j\}_1^\infty \subset \mathcal{A}$ and $E_1 \subset E_2 \subset \dots$, then $\bigcup_1^\infty E_j \in \mathcal{A}$).

Proof. If $\mathcal{A}$ is a σ -algebra, then $\mathcal{A}$ is closed under countable unions in general, so it is certainly closed under countable increasing unions in particular. On the other hand, if $\mathcal{A}$ is an algebra which is closed under countable increasing unions, then take any $\{E_j\}_1^\infty \subset \mathcal{A}$. Consider that

$$\bigcup_{j=1}^\infty E_j = \bigcup_{j=1}^\infty \bigcup_{k=1}^j E_k. \quad (1)$$

Clearly, for any $j \in \mathbb{N}$, $\left(\bigcup_{k=1}^j E_k\right) \subset \left(\bigcup_{k=1}^{j+1} E_k\right)$, and each $\bigcup_{k=1}^j E_k$ is a finite union of elements of the algebra $\mathcal{A}$, and thus is in $\mathcal{A}$. Then the right hand side of (1) is a countable increasing union of members of $\mathcal{A}$. Thus (1) $\in \mathcal{A}$. □

Exercise 5. If $\mathcal{M}$ is the σ -algebra generated by $\mathcal{E}$, then $\mathcal{M}$ is the union of the σ -algebras generated by $\mathcal{F}$ as $\mathcal{F}$ ranges over all countable subsets of $\mathcal{E}$.

Proof. Let F be the set of all countable subsets of $\mathcal{E}$. By Lemma 1.1, since $\mathcal{F} \subset \mathcal{E} \subset \mathcal{M}(\mathcal{E})$ for each $\mathcal{F} \in F$, $\mathcal{M}(\mathcal{F}) \subset \mathcal{M}(\mathcal{E})$. Their union, then, is also a subset of $\mathcal{E}$.

We need to show, then, that $\mathcal{M}(\mathcal{E}) \subset \bigcup_{\mathcal{F} \in F} \mathcal{M}(\mathcal{F})$. If we can show that $\bigcup_{\mathcal{F} \in F} \mathcal{M}(\mathcal{F}) = \mathcal{M}(\bigcup_{\mathcal{F} \in F} \mathcal{F})$, then we will be done because $\mathcal{E} \subset \bigcup F$, so $\mathcal{M}(\mathcal{E}) \subset \mathcal{M}(\bigcup F)$. That is where we now turn our focus.

Clearly $\mathcal{M}(\bigcup F) \subset \bigcup_{\mathcal{F} \in F} \mathcal{M}(\mathcal{F})$ since $\bigcup F \subset \bigcup_{\mathcal{F} \in F} \mathcal{M}(\mathcal{F})$, so we need to prove the other direction. Given $E \in \bigcup \mathcal{M}(F)$, fix some $\mathcal{F} \in F$ for which $E \in \mathcal{M}(\mathcal{F})$. Then since $\mathcal{F} \subset \bigcup F$, $\mathcal{M}(\mathcal{F}) \subset \mathcal{M}(\bigcup F)$, so $E \in \mathcal{M}(\bigcup F)$. As observed above, this completes our proof. □